THE MINIMUM NUMBER OF DISTINCT ABELIAN SQUARES IN A BINARY CIRCULAR WORD

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Abstract. We prove that every binary circular word of length n contains at least $\lfloor (n-1)/2 \rfloor$ distinct abelian squares, and we determine all words attaining this bound. This settles a conjecture of Simpson: its odd case holds, and in its even case the list of extremal words misses $(ab)^{n/2}$ for $n \equiv 2 \pmod 4$, $n \geq 6$. We then review the other problems of the survey: we conjecture, and verify to length 27, that for odd n the least number of non-equivalent abelian squares in a circular word is also $(n-1)/2$, and we show that the least number of occurrences of abelian squares in a binary word of length n is asymptotically proportional to n or grows faster, and is below $n^{3/2}$.

Note. An earlier version of this paper appeared on Hexagon, where it is the preprint `hexagon:2610.00473`, available at `https://hexagonmath.org/2610.00473v1`. That version proved Theorem 1.1 and stopped there. This version adds Sections 5 and 6, which treat the other problems of Simpson's survey: the balance identities, the jump and descent lemmas, the two-run formula and Conjecture 5.5 on non-equivalent abelian squares in circular words of odd length, verified to length 27; the status of the linear-word minimum $\lfloor n/4 \rfloor$ and of the problems of Fici and Mignosi, of Kucherov, Ochem and Rao and of Mäkelä; and Theorem 6.1 on the least number of occurrences of abelian squares. The earlier versions are archived on Zenodo under the concept DOI 10.5281/zenodo.23313258 (versions 10.5281/zenodo.23313259, 10.5281/zenodo.23313541, 10.5281/zenodo.23384530, 10.5281/zenodo.23553002 and 10.5281/zenodo.23672660). The source of this revision, the Lean formalization and the programs are in the repository `https://github.com/creelie/literate-octo-lamp`, release `v2.3.0`, archived on Zenodo as 10.5281/zenodo.23692500.

1. Introduction

Words are over the alphabet $\{a, b\}$. For a letter ℓ we write $\bar{\ell}$ for the other letter and $|u|_\ell$ for the number of occurrences of ℓ in a word u. An *abelian square* is a word uv with u nonempty and $|u|_a = |v|_a$, $|u|_b = |v|_b$; then $|u| = |v|$. A circular word of length n is a word $w = w_0 \cdots w_{n-1}$ read cyclically; its

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factors are the words $w_i w_{i+1} \cdots w_{i+L-1}$, indices modulo n, with $0 \le i < n$ and $1 \le L \le n$. As in [1], factors longer than n are not counted. Let $D(w)$ be the number of distinct factors of w that are abelian squares. Conjugate words (cyclic shifts) and complementary words (obtained by exchanging a and b) have the same value of D. Simpson [1] surveyed extremal problems on abelian squares in linear and circular words; for the minimum number of distinct abelian squares in a circular word he stated the following, based on computer experiments.

Conjecture 0.6 of [1]**.** The minimum number of distinct abelian squares in a circular word of length n is:

(a) $(n-1)/2$ if n is odd and this bound is attained only by a^n, $a^{n-1}b$ and their complements and conjugates.
(b) $(n-2)/2$ if n is even and this bound is attained only by $a^k b^{n-k}$ and its complement and their conjugates, where $k \in \{1, 3, 5, \ldots, n-1\}$.

For non-equivalent abelian squares in circular words Fraenkel, Simpson and Paterson [2] found the minimum when $n \equiv 2 \pmod 4$; the odd case is discussed in Section 5. For linear binary words the minimum number of distinct abelian squares is conjectured by Fici and Saarela to be $\lfloor n/4 \rfloor$, and [3] proves this in special cases; maximal numbers of abelian squares were studied in [4, 5], and [6] surveys the area. We prove the following theorem; the remaining problems of the survey are discussed in Sections 5 and 6.

Theorem 1.1. *Let $n \ge 1$. Every binary circular word w of length n satisfies $D(w) \ge \lfloor (n-1)/2 \rfloor$. Equality holds if and only if w is a conjugate of one of the following words:*

(a) *n odd: a^n, b^n, $a^{n-1}b$, $b^{n-1}a$;*
(b) *n even: $a^k b^{n-k}$ with k odd, and $(ab)^{n/2}$ if $n \equiv 2 \pmod 4$.*

The complement $b^k a^{n-k}$ of $a^k b^{n-k}$ is a conjugate of $a^{n-k}b^k$, and $n-k$ is odd when n is even and k is odd; the complement of $(ab)^{n/2}$ is a conjugate of $(ab)^{n/2}$. Hence the lists in (a) and (b) are closed under complement and conjugation.

Corollary 1.2. *Part (a) of Conjecture 0.6 of* [1] *holds. In part (b) the minimum $(n-2)/2$ is correct, and the list of extremal words is complete if and only if $n \not\equiv 2 \pmod 4$ or $n = 2$. For $n \equiv 2 \pmod 4$, $n \ge 6$, the words $(ab)^{n/2}$ and $(ba)^{n/2}$ attain the minimum and are not conjugates of $a^k b^{n-k}$ or $b^k a^{n-k}$ for any k.*

Proof. For $n \ge 4$ the word $(ab)^{n/2}$ has $n/2 \ge 2$ maximal blocks of a, while every conjugate of $a^k b^{n-k}$ or $b^k a^{n-k}$ has at most one maximal block of a when read cyclically. For $n = 2$, $(ab)^1 = a^1 b^1$. The rest is Theorem 1.1. □

For example, the abelian squares among the factors of the circular word *ababab* are *abab* and *baba*, so $D(ababab) = 2 = (6-2)/2$. Counting conjugates, the number of words of length n with $D(w) = \lfloor (n-1)/2 \rfloor$ is $2n+2$ for odd $n \ge 3$, and $n^2/2$ for even n, plus 2 when $n \equiv 2 \pmod 4$ and $n \ge 6$. Figure 1 shows the extremal words of length 6 up to conjugation and complement.

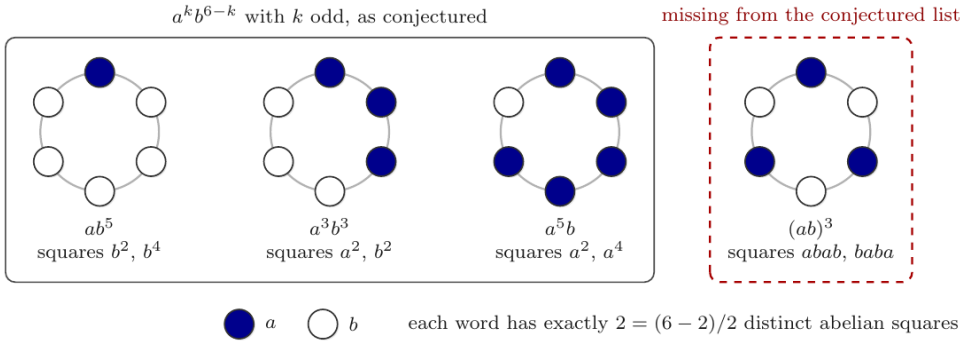


FIGURE 1. The extremal circular words of length 6, read clockwise from the top, up to complement. The three words on the left are those of Conjecture 0.6(b); the alternating word on the right also has exactly two distinct abelian squares.

Section 2 proves the lower bound by a double count of the abelian squares of half-length m and $n-m$. Section 3 describes the words in which this count is tight for one letter, and Section 4 completes the proof. Section 7 describes the Lean 4 formalization.

2. The lower bound

Fix a circular word w of length $n \geq 1$ and set $x_i = w_{i \bmod n}$ for $i \in \mathbb{Z}$. For a letter ℓ let $t_\ell = |w|_\ell$, so $t_a + t_b = n$. For $i \in \mathbb{Z}$ and $L \geq 0$ let $c_\ell(i, L)$ be the number of $s \in [i, i+L)$ with $x_s = \ell$; a window of n consecutive positions contains t_ℓ occurrences of ℓ, so $c_\ell(i, L) < t_\ell$ implies $L < n$. For $m \geq 1$ let $W(i, m) = x_i x_{i+1} \cdots x_{i+2m-1}$; we call $W(i, m)$ a *square* if $c_a(i, m) = c_a(i+m, m)$, equivalently $c_b(i, m) = c_b(i+m, m)$.

A factor of w that is an abelian square has even length $2m \leq n$, and it equals $W(i, m)$ for some i for which $W(i, m)$ is a square. Hence

$$D(w) = \sum_{m=1}^{\lfloor n/2 \rfloor} |S_m|, \qquad S_m = \{W(i, m) : i \in \mathbb{Z},\ W(i, m) \text{ is a square}\}.$$

Definition 2.1. For $1 \leq m \leq n-1$ let Q_m be the set of pairs (ℓ, k) such that some square $W(i, m)$ satisfies $x_{i+m-1} = \ell$ and $c_\ell(i, m) = k$. For a letter ℓ and $k \geq 0$, the *column* $C_\ell(k)$ is the set of $m \in [1, n-1]$ with $(\ell, k) \in Q_m$.

So $(\ell, k) \in Q_m$ records a square of half-length m whose left half ends with ℓ and contains k letters ℓ. The sets S_m, Q_m and $C_\ell(k)$ do not change when w is replaced by a conjugate.

Lemma 2.2. *$|Q_m| \leq |S_m|$ for $1 \leq m \leq n/2$.*

Proof. The pair (ℓ, k) attached to a square $W(i, m)$ is determined by the word $W(i, m)$: ℓ is its m-th letter and k the number of letters ℓ among its first m letters. So Q_m is the image of S_m under a map. □

Lemma 2.3. *Let $1 \leq m \leq n-1$. If $(\ell, k) \in Q_m$, then $(\ell, t_\ell - k) \in Q_{n-m}$. Hence $|Q_m| = |Q_{n-m}|$, and $1 \leq k \leq t_\ell - 1$ for every $(\ell, k) \in Q_m$.*

Proof. Let $W(i,m)$ be a square with $x_{i+m-1} = \ell$ and $c_\ell(i,m) = c_\ell(i+m,m) = k$. Consider $W(i+2m, n-m)$. Its left half $[i+2m, i+m+n)$ and the right half $[i+m, i+2m)$ of $W(i,m)$ together form the window $[i+m, i+m+n)$, so the left half contains $t_\ell - k$ letters ℓ. Its right half $[i+m+n, i+2n)$ is the translate by n of $[i+m, i+n)$, which together with $[i, i+m)$ forms a window of length n; so it also contains $t_\ell - k$ letters ℓ. The last letter of the left half is $x_{i+m+n-1} = x_{i+m-1} = \ell$. Thus $(\ell, t_\ell - k) \in Q_{n-m}$. The map $(\ell, k) \mapsto (\ell, t_\ell - k)$ is injective and is its own inverse, so $|Q_m| = |Q_{n-m}|$. Finally $k \geq 1$ because the left half ends with ℓ, and $t_\ell - k \geq 1$ by the same argument applied to $W(i+2m, n-m)$. □

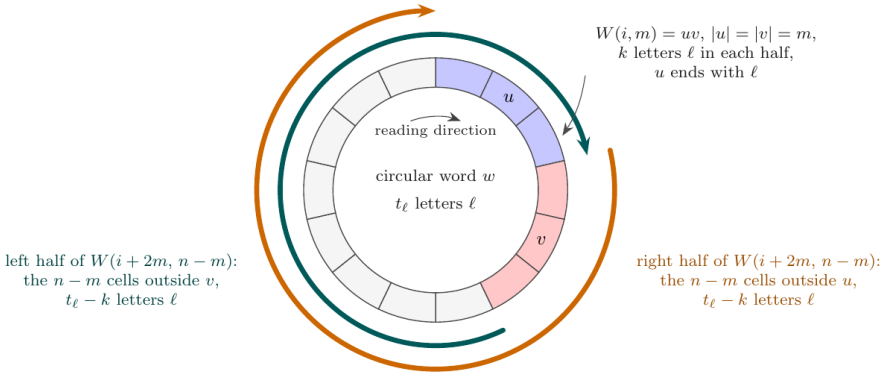


FIGURE 2. The proof of Lemma 2.3. The two halves of the square $W(i+2m, n-m)$ are the complements of v and of u in the circle, so each contains $t_\ell - k$ letters ℓ, and the left one ends at the last letter of u.

If $t_\ell \geq 1$, let $(p_j)_{j \in \mathbb{Z}}$ be the increasing enumeration of $\{s \in \mathbb{Z} : x_s = \ell\}$; then $p_{j+t_\ell} = p_j + n$. For $j \in \mathbb{Z}$ and $1 \leq k < t_\ell$ let $I_j(k)$ be the interval of integers m with

$$\max(p_{j+k} - p_{j+1} + 1,\ p_{j+2k} - p_{j+k}) \ \leq\ m \ \leq\ \min(p_{j+k} - p_j,\ p_{j+2k+1} - p_{j+k} - 1). \tag{1}$$

Lemma 2.4. *Let $1 \leq k < t_\ell$ and $m \in I_j(k)$. Then $W(p_{j+k} + 1 - m, m)$ is a square whose left half ends with ℓ and contains k letters ℓ. In particular $m \in C_\ell(k)$.*

Proof. Let $i = p_{j+k} + 1 - m$. By (1), $p_j < i \leq p_{j+1}$, so the left half $[i, p_{j+k}]$ contains exactly the occurrences $p_{j+1}, \ldots, p_{j+k}$ and ends with $x_{p_{j+k}} = \ell$. Also $p_{j+2k} \leq p_{j+k} + m < p_{j+2k+1}$, so the right half $[p_{j+k} + 1, p_{j+k} + m]$ contains exactly $p_{j+k+1}, \ldots, p_{j+2k}$. Both halves contain k letters ℓ. Since $m \geq 1$ and $k < t_\ell$, we have $1 \leq m < n$. □

Lemma 2.5. *For every letter ℓ and $1 \leq k \leq t_\ell - 1$, the column $C_\ell(k)$ is nonempty.*

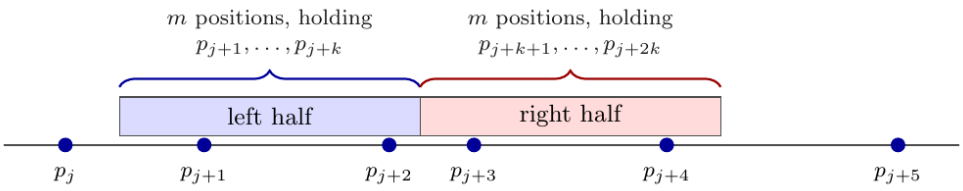


FIGURE 3. The square of Lemma 2.4 for $k = 2$. The occurrences of ℓ are the dots; the left half ends at p_{j+k} and the right half stops before p_{j+2k+1}.

Proof. Let $d_j = p_{j+k} - p_j$ and $e_j = d_{j+k} - d_j$. Both are periodic in j with period t_ℓ, so $\sum_{j=r}^{r+t_\ell-1} e_j = 0$ for every r. Hence some $e_{j_0} \le 0$. If $e_j \le 0$ implied $e_{j+1} < 0$ for every j, then $e_j < 0$ for all $j > j_0$, and $\sum_{j=j_0+1}^{j_0+t_\ell} e_j < 0$. So there is j with $e_j \le 0 \le e_{j+1}$, that is $d_{j+k} \le d_j$ and $d_{j+1} \le d_{j+k+1}$. We show $I_j(k) \ne \emptyset$ by comparing each lower bound in (1) with each upper bound. First, $p_{j+k} - p_{j+1} + 1 \le p_{j+k} - p_j$ and $p_{j+2k} - p_{j+k} \le p_{j+2k+1} - p_{j+k} - 1$. Second, $p_{j+2k} - p_{j+k} = d_{j+k} \le d_j = p_{j+k} - p_j$. Third, with $g = p_{j+k+1} - p_{j+k} \ge 1$,

$$\begin{aligned} p_{j+k} - p_{j+1} + 1 = d_{j+1} - g + 1 &\le d_{j+1} \le d_{j+k+1} \\ &\le d_{j+k+1} + g - 1 = p_{j+2k+1} - p_{j+k} - 1. \end{aligned}$$

By Lemma 2.4, $C_\ell(k) \supseteq I_j(k) \ne \emptyset$. □

Theorem 2.6. $D(w) \ge \lfloor (n-1)/2 \rfloor$.

Proof. Let $N = \sum_{m=1}^{n-1} |Q_m|$ and $F = \sum_{m=1}^{\lfloor n/2 \rfloor} |Q_m|$. Counting the pairs $(m, (\ell, k))$ with $(\ell, k) \in Q_m$ by columns,

$$N = \sum_{\ell} \sum_{k \ge 0} |C_\ell(k)|, \qquad \sum_{k \ge 0} |C_\ell(k)| \ge t_\ell - 1 \ \text{ if } t_\ell \ge 1, \tag{2}$$

by Lemmas 2.3 and 2.5. If both letters occur, $N \ge t_a + t_b - 2 = n - 2$; otherwise $N \ge n - 1$. By Lemma 2.3, $|Q_m| = |Q_{n-m}|$, so

$$N + \varepsilon\, |Q_{n/2}| = 2F, \qquad \varepsilon = 1 \text{ if } n \text{ is even, } \varepsilon = 0 \text{ if } n \text{ is odd.} \tag{3}$$

Hence $2F \ge n - 2$ and $F \ge \lceil (n-2)/2 \rceil = \lfloor (n-1)/2 \rfloor$. By Lemma 2.2, $D(w) \ge F$. □

We call a letter ℓ with $t_\ell \ge 1$ *tight* if $|C_\ell(k)| \le 1$ for every k. By Lemmas 2.3 and 2.5, ℓ is tight if and only if $\sum_k |C_\ell(k)| = t_\ell - 1$. We call w *rigid* if $|S_m| = |Q_m|$ for $1 \le m \le n/2$: two squares $W(i, m)$, $W(i', m)$ with $2m \le n$, $x_{i+m-1} = x_{i'+m-1} = \ell$ and $c_\ell(i, m) = c_\ell(i', m)$ are equal words.

Proposition 2.7. *If $D(w) = \lfloor (n-1)/2 \rfloor$, then w is rigid and $N + \varepsilon |Q_{n/2}| = 2\lfloor (n-1)/2 \rfloor$.*

Proof. In the proof of Theorem 2.6, $\lfloor (n-1)/2 \rfloor \le F \le D(w)$, so $F = D(w)$, and $|Q_m| \le |S_m|$ for each m forces $|Q_m| = |S_m|$. Then (3) gives the second statement. □

3. Rigid words with a tight letter

Theorem 3.1. *Let w be rigid and let ℓ be a tight letter, $t = t_\ell$. Then w is a conjugate of $\ell^t\bar{\ell}^{n-t}$ with t odd or $t = n$, or $n = 2t$ and w is a conjugate of $(\ell\bar{\ell})^t$.*

Throughout this section w and ℓ are as in Theorem 3.1, (p_j) enumerates the occurrences of ℓ, and $g_j = p_{j+1} - p_j \geq 1$. Since ℓ is tight, Lemma 2.4 gives:

$$\text{for } 1 \leq k < t, \text{ the integers in } \bigcup_j I_j(k) \text{ form a set with at most one element.} \tag{4}$$

Proof of Theorem 3.1. If $t = n$, then $w = \ell^n$. Assume $t < n$, so $g_j \geq 2$ for some j.

Case A: $g_j \geq 2$ for all j. If $t = 1$, then w is a conjugate of $\ell\bar{\ell}^{n-1}$. Let $t \geq 2$. By (1), $I_j(1)$ is the set of m with $g_{j+1} \leq m \leq \min(g_j,\, g_{j+1} + g_{j+2} - 1)$. If $g_j > g_{j+1}$, then $I_j(1) \supseteq \{g_{j+1}, g_{j+1} + 1\}$ since $g_{j+2} \geq 2$, contradicting (4). So $g_j \leq g_{j+1}$ for all j, and as g is periodic, $g_j = \delta$ is constant and $n = t\delta$. If $\delta = 2$, then $n = 2t$ and w is a conjugate of $(\ell\bar{\ell})^t$. Let $\delta \geq 3$, so $p_j = p_0 + j\delta$. The words $W(p_1, \delta)$ and $W(p_1 + 2, \delta)$ have halves $\ell\bar{\ell}^{\delta-1}$, $\ell\bar{\ell}^{\delta-1}$ and $\bar{\ell}^{\delta-2}\ell\bar{\ell}$, $\bar{\ell}^{\delta-2}\ell\bar{\ell}$. Both are squares whose left halves end with $\bar{\ell}$ and contain $\delta - 1$ letters $\bar{\ell}$, and $2\delta \leq n$. By rigidity they are equal, but their first letters are ℓ and $\bar{\ell}$.

Case B: $g_j = 1$ for some j. A *run* of length $L \geq 1$ consists of occurrences $p_{J+1}, \ldots, p_{J+L}$ with $g_J \geq 2$, $g_{J+1} = \cdots = g_{J+L-1} = 1$ and $g_{J+L} \geq 2$; its *gaps* are $\lambda = g_J$ and $\rho = g_{J+L}$. Since g has period t and some $g_j \geq 2$, every run has length $L \leq t$. The two occurrences p_j, p_{j+1} with $g_j = 1$ lie in a run of length at least 2.

(B1) *Every run has odd length.* Suppose a run has length $2i$, $i \geq 1$. In (1) for $I_J(i)$ the lower bounds are i and i, and the upper bounds are $\lambda + i - 1 \geq i + 1$ and $i + \rho - 1 \geq i + 1$. So $I_J(i) \supseteq \{i, i + 1\}$ with $i < 2i \leq t$, contradicting (4).

For a run $p_{J+1}, \ldots, p_{J+2h+1}$ with $h \geq 1$ and gaps λ, ρ, (1) gives, with $\lambda' = g_{J-1}$ and $\rho' = g_{J+2h+2}$,

$$I_{J-1}(h+1) = [\lambda + h,\ \min(\lambda + h + \lambda' - 1,\ h + \rho)], \tag{5}$$

$$I_J(h+1) = [\rho + h,\ \min(\lambda + h,\ h + \rho + \rho' - 1)], \tag{6}$$

where we used $\lambda, \rho \geq 2$. Here $h + 1 < 2h + 1 \leq t$.

(B2) *All runs of length at least 3 have the same length.* Let a run $p_{J+1}, \ldots, p_{J+2h+1}$ have $h \geq 1$, and suppose another run $p_{J'+1}, \ldots, p_{J'+L'}$ has $L' \geq 2h + 2$. In (1) for $I_{J'}(h+1)$ the lower bounds are $h + 1$ and $h + 1$, and the upper bounds are at least $h + 1$; so $h + 1 \in I_{J'}(h + 1)$. By (5) and (6), $\lambda + h \in I_{J-1}(h + 1)$ if $\lambda \leq \rho$ and $\rho + h \in I_J(h + 1)$ if $\rho \leq \lambda$. Both numbers exceed $h + 1$, contradicting (4).

By (B1) and (B2) there is $h \geq 1$ such that every run has length 1 or $2h + 1$, and some run has length $2h + 1$. Consider such a run $R = (p_{J+1}, \ldots, p_{J+2h+1})$ with gaps λ, ρ.

(B3) *$\rho < \lambda$ is impossible.* Let R' be the run following R, starting at p_{J+2h+2}. If R' has length 1, then $\rho' \geq 2$, and by (6) $I_J(h+1) \supseteq \{\rho + h, \rho + h + 1\}$. If R' has length $2h + 1$, let $\sigma \geq 2$ be its right gap. Then $t \neq 2h + 1$: otherwise R' is the translate of R by n and $\rho = g_{J+t} = g_J = \lambda$. In (1) for $I_J(2h+1)$ the lower

bounds are $2h+1$ and $2h+\rho$, and the upper bounds are $2h+\lambda \geq 2h+\rho+1$ and $2h+\rho+\sigma-1 \geq 2h+\rho+1$; so $I_J(2h+1) \supseteq \{2h+\rho, 2h+\rho+1\}$ with $2h+1 < t$. Both cases contradict (4).

(B4) *If $\lambda < \rho$, the occurrence p_J is not a run of length* 1. Otherwise $\lambda' \geq 2$, and by (5) $I_{J-1}(h+1) \supseteq \{\lambda+h, \lambda+h+1\}$, contradicting (4).

(B5) *If $\lambda = \rho$, then $t = 2h+1$ and w is a conjugate of $\ell^{2h+1}\bar{\ell}^{\lambda-1}$.* By (5) and (6), $m = \lambda+h$ lies in $I_{J-1}(h+1)$ and in $I_J(h+1)$. By Lemma 2.4 the squares $W(p_J, m)$ and $W(p_J+1, m)$ have left halves ending with ℓ and containing $h+1$ letters ℓ. Their first letters are $x_{p_J} = \ell$ and $x_{p_J+1} = \bar{\ell}$, since $p_J < p_J+1 < p_{J+1}$. By rigidity, $2m > n$. Then $p_{J+t} = p_J + n < p_J + 2\lambda + 2h = p_{J+2h+2}$, so $t \leq 2h+1$, and $t \geq 2h+1$ as R has $2h+1$ occurrences. So $t = 2h+1$, $n = p_{J+2h+2} - p_{J+1} = 2h+\lambda$, and $x_{p_{J+1}} \cdots x_{p_{J+1}+n-1} = \ell^{2h+1}\bar{\ell}^{\lambda-1}$.

Now choose the run R of length $2h+1$ with the least left gap λ. By (B3), $\rho \geq \lambda$. If $\rho = \lambda$, (B5) applies. If $\rho > \lambda$, then by (B4) the run R^- preceding R has length $2h+1$. Its right gap is λ, and its left gap is at least λ by the choice of R; by (B3) the two are equal, and (B5) applies to R^-. In all cases w is a conjugate of $\ell^t\bar{\ell}^{n-t}$ with $t = 2h+1$ odd. □

4. Proof of Theorem 1.1

Lemma 4.1. *Let ℓ be a letter.*

(i) *Let $0 \leq \alpha \leq n$ and $\beta = n-\alpha$, and suppose each of α, β is 0, odd, or at least $n-1$. Then every abelian square factor of the circular word $\ell^\alpha\bar{\ell}^\beta$ is ℓ^{2m} or $\bar{\ell}^{2m}$, and $D(\ell^\alpha\bar{\ell}^\beta) = \lfloor\alpha/2\rfloor + \lfloor\beta/2\rfloor$.*

(ii) *For even n, $D((\ell\bar{\ell})^{n/2}) = 2\lfloor n/4\rfloor$.*

Proof. (i) Let $s = uv$ be an abelian square factor, $|u| = |v| = m$, $2m \leq n$, containing both letters; then $1 \leq \alpha, \beta \leq n-1$, and read cyclically the word consists of the two blocks ℓ^α and $\bar{\ell}^\beta$. If s crosses exactly one block boundary, $s = c^r\bar{c}^{2m-r}$ with $c \in \{\ell, \bar{\ell}\}$ and $0 < r < 2m$, and $|u|_c = \min(r, m) > \max(r-m, 0) = |v|_c$. Otherwise s contains a whole block in its interior, say $s = \ell^r\bar{\ell}^\beta\ell^{r'}$ with $r, r' \geq 1$. Then $\beta \leq 2m-2 \leq n-2$, so β is odd, whereas $\beta = |s|_{\bar{\ell}} = 2|u|_{\bar{\ell}}$. The case $s = \bar{\ell}^r\ell^\alpha\bar{\ell}^{r'}$ is symmetric. The factor ℓ^{2m} occurs if and only if $2m \leq \alpha$, and $\bar{\ell}^{2m}$ if and only if $2m \leq \beta$.

(ii) Every factor of length $2m$ alternates. If m is odd, its halves start with different letters $c \neq \bar{c}$, and $|u|_c = (m+1)/2 \neq (m-1)/2 = |v|_c$. If m is even, both halves contain $m/2$ letters of each kind, and there are exactly two factors of length $2m \leq n$. So $D = 2\,|\{m : m \text{ even},\ 2 \leq m \leq n/2\}| = 2\lfloor n/4\rfloor$. □

Lemma 4.2. *Let $\alpha \geq 3$, $h \geq 1$, $n = \alpha + 2h$ and $w = \ell^\alpha\bar{\ell}^{2h}$. Then $C_{\bar{\ell}}(h) \supseteq \{h,\ h+1,\ n-h\}$, three distinct numbers.*

Proof. $W(\alpha, h) = \bar{\ell}^{2h}$ is a square whose left half $\bar{\ell}^h$ ends with $\bar{\ell}$, so $h \in C_{\bar{\ell}}(h)$. $W(\alpha-1, h+1) = \ell\bar{\ell}^{2h}\ell$ is a square with halves $\ell\bar{\ell}^h$ and $\bar{\ell}^h\ell$, and $2h+2 \leq n$; so $h+1 \in C_{\bar{\ell}}(h)$. By Lemma 2.3 with $t_{\bar{\ell}} = 2h$, $n-h \in C_{\bar{\ell}}(2h-h)$. Finally $n-h = \alpha+h \geq h+3$. □

Proof of Theorem 1.1. The lower bound is Theorem 2.6. If w is a conjugate of a word in the list, Lemma 4.1 gives $D(w) = \lfloor(n-1)/2\rfloor$: for a^n (n odd) $\lfloor n/2\rfloor = (n-1)/2$; for $a^{n-1}b$ (n odd) $(n-1)/2+0$; for a^kb^{n-k} (n even, k odd)

$(k-1)/2+(n-k-1)/2=(n-2)/2$; for $(ab)^{n/2}$ ($n\equiv 2 \pmod 4$) $2(n-2)/4$. Complements and conjugates have the same D.

Conversely let $D(w)=\lfloor (n-1)/2\rfloor$. By Proposition 2.7, w is rigid and $N+\varepsilon|Q_{n/2}|=2\lfloor (n-1)/2\rfloor$.

n even. Then $N+|Q_{n/2}|=n-2$. If only one letter occurs, $N\geq n-1$; so both occur, and by (2) $N\geq n-2$. Hence $|Q_{n/2}|=0$ and $\sum_k |C_\ell(k)|=t_\ell-1$ for both letters, so both letters are tight. Theorem 3.1 for $\ell=a$ shows that w is a conjugate of a^kb^{n-k} with k odd ($k=n$ is excluded since b occurs), or $n=2t_a$ and w is a conjugate of $(ab)^{n/2}$. In the second case, if $n\equiv 0 \pmod 4$, both halves of $W(0,n/2)$ alternate and have even length $n/2$, so $W(0,n/2)$ is a square and $Q_{n/2}\neq\emptyset$, a contradiction. So $n\equiv 2 \pmod 4$.

n odd. Then $N=n-1$. If only one letter occurs, w is a^n or b^n. Otherwise (2) gives $\sum_k |C_a(k)|+\sum_k |C_b(k)|=(t_a-1)+(t_b-1)+1$ with $\sum_k |C_\ell(k)|\geq t_\ell-1$ for both letters. So one letter ℓ is tight, and $\sum_k |C_{\bar\ell}(k)|\leq t_{\bar\ell}$. By Theorem 3.1, w is a conjugate of $\ell^t\bar\ell^{n-t}$ with $t=t_\ell$ odd, since $t<n$ and n is odd. Then $n-t=2h$ with $h\geq 1$. If $t=1$, then w is a conjugate of $\bar\ell^{n-1}\ell$. If $t\geq 3$, Lemma 4.2 gives $|C_{\bar\ell}(h)|\geq 3$ with $1\leq h<2h=t_{\bar\ell}$, and with Lemma 2.5, $\sum_k |C_{\bar\ell}(k)|\geq (t_{\bar\ell}-1)+2>t_{\bar\ell}$, a contradiction. □

5. Non-equivalent abelian squares in circular words of odd length

Two abelian squares are *equivalent* [2, 1] when they have the same length and the same number of letters a. The *type* of a square $W(i,m)$ is the pair $(m,c_a(i,m))$, and for $1\leq m\leq n/2$ we let $z_m(w)$ be the number of types of the squares $W(i,m)$, $i\in\mathbb{Z}$. Thus

$$N(w)=\sum_{m=1}^{\lfloor n/2\rfloor} z_m(w)$$

is the number of non-equivalent abelian squares among the factors of w, and $N(w)\leq D(w)$ since the type of a square is determined by the factor. Fraenkel, Simpson and Paterson [2] proved that a circular word with at most k non-equivalent abelian squares has length at most $4k+2$, the bound being attained by $(ab)^{2k+1}$ only; hence $N(w)\geq\lfloor (n+1)/4\rfloor$ for every circular word of length n, and this is sharp when $n\equiv 2 \pmod 4$. In this section n is odd. Then $N(w)\leq D(w)$ and Theorem 1.1 give $N(w)\leq (n-1)/2$ for the words a^n and $a^{n-1}b$, so the minimum of N over the words of odd length n is at most $(n-1)/2$. We collect what we can prove about the lower bound and state the expected answer as a conjecture; all the computations referred to are performed by the programs `odd_circular_types.c` and `linear_minima.c` of the ancillary files.

For $i\in\mathbb{Z}$ and $1\leq m\leq n-1$ put $\delta_m(i)=c_a(i+m,m)-c_a(i,m)$, so that $W(i,m)$ is a square exactly when $\delta_m(i)=0$. Since $i\mapsto i+m$ permutes the residues modulo n,

$$\sum_{i=0}^{n-1}\delta_m(i)=0, \tag{7}$$

and writing $[x_s=a]$ for 1 if $x_s=a$ and 0 otherwise,

$$\delta_m(i+1)-\delta_m(i)=[x_i=a]-2[x_{i+m}=a]+[x_{i+2m}=a]\in\{-2,-1,0,1,2\}. \tag{8}$$

Lemma 5.1 (jump)**.** *Let $1 \le m$ with $2m < n$, and suppose that no $W(i,m)$ is a square. Then there are i and i' with*

$$x_i = x_{i+2m} = a, \quad x_{i+m} = b, \quad c_a(i+1,m) = c_a(i+m,m),$$
$$x_{i'} = x_{i'+2m} = b, \quad x_{i'+m} = a, \quad c_a(i'+1,m) = c_a(i'+m,m).$$

Proof. By assumption δ_m has no zero, and by (7) it takes a negative and a positive value. Going once around the circle we find i with $\delta_m(i) < 0 < \delta_m(i+1)$ and i' with $\delta_m(i') > 0 > \delta_m(i'+1)$. At i the step (8) is at least 2, so it equals 2, $\delta_m(i) = -1$, $\delta_m(i+1) = 1$, $x_i = x_{i+2m} = a$ and $x_{i+m} = b$. Then $c_a(i+1,m) = c_a(i,m) - [x_i = a] + [x_{i+m} = a] = c_a(i,m) - 1 = c_a(i+m,m)$. The position i' is treated in the same way, with the step equal to -2: $x_{i'} = x_{i'+2m} = b$, $x_{i'+m} = a$, $\delta_m(i') = 1$ and $c_a(i'+1,m) = c_a(i',m) + 1 = c_a(i'+m,m)$. □

In words: an empty half-length m forces the two patterns $aXbYa$ and $bX'aY'b$ in w, with $|X| = |Y| = |X'| = |Y'| = m-1$, $|X|_a = |Y|_a$ and $|X'|_a = |Y'|_a$. Such a pattern produces squares of smaller half-lengths.

Lemma 5.2 (descent)**.** *Let $m \ge 2$ and let i satisfy $x_i = x_{i+2m} = a$, $x_{i+m} = b$ and $c_a(i+1,m-1) = c_a(i+m+1,m-1) = e$. Then either $(m-1,e)$ is the type of a square, or $x_{i+1} = x_{i+2m-1} = a$ and $i+1$ satisfies the same hypotheses with $m-1$ and $e-1$ in place of m and e. The statement with the letters a and b exchanged, and e counting the letters b, holds as well.*

Proof. If $x_{i+2m-1} = b$ then $W(i+1,m-1)$ has halves $x_{i+1}\cdots x_{i+m-1}$ and $x_{i+m}\cdots x_{i+2m-2}$ with e and $c_a(i+m+1,m-1) - [x_{i+2m-1} = a] = e$ letters a (recall $x_{i+m} = b$), so it is a square of type $(m-1,e)$. If $x_{i+1} = b$ then $W(i+2,m-1)$ has halves $x_{i+2}\cdots x_{i+m}$ and $x_{i+m+1}\cdots x_{i+2m-1}$ with $e - [x_{i+1} = a] = e$ and e letters a, again a square of type $(m-1,e)$. Otherwise $x_{i+1} = x_{i+2m-1} = a$; then $x_{i+1} = a$, $x_{(i+1)+(m-1)} = x_{i+m} = b$, $x_{(i+1)+2(m-1)} = x_{i+2m-1} = a$, and $c_a(i+2,m-2) = e - 1 = c_a(i+m+1,m-2)$ because the first window lost the letter $x_{i+1} = a$ and the second lost $x_{i+2m-1} = a$. The second statement follows by complementation. □

Starting from the pattern of Lemma 5.1 and applying Lemma 5.2 repeatedly, an empty half-length m yields a square of some type $(m-1-r, e-r)$ with $0 \le r \le m-2$, or else the factor $a^m b a^m$, which contains the squares a^{2j} for $2j \le m$; and likewise with b. The squares so obtained need not be new, and this is why the following two propositions cover only the smallest half-lengths and the words with two runs. Here a *run* is a maximal factor of the circular word consisting of one letter.

Proposition 5.3. *Let $n \ge 5$ be odd. Then $z_1(w) \ge 1$ and $z_1(w) + z_2(w) \ge 2$ for every circular word w of length n.*

Proof. Since n is odd, w is not a conjugate of $(ab)^{n/2}$, so $x_i = x_{i+1}$ for some i and $W(i,1)$ is a square: $z_1(w) \ge 1$. Suppose $z_1(w) = 1$; by complementation we may assume that bb is not a factor, so each run of b has length 1. If $t_b = 0$ then $w = a^n$, $n \ge 5$, and $aaaa$ gives $z_2(w) \ge 1$. Let $t_b \ge 1$. A run of a of length at least 4 gives the square $aaaa$ of type $(2,2)$. A run of a of length exactly 2 lies between two letters b, which gives the factor $baab$, a square of type $(2,1)$. A run of a of length 1 is a factor bab, and since the letter after

the second b is a (if $t_b = 1$ this second b is the first one and $n = 2$, which is excluded), w has the factor $baba$, a square of type $(2, 1)$. If none of these happens, every run of a has length 3 and $n = 4t_b$ is even, a contradiction. □

Proposition 5.4. *Let $n = p + q$ be odd with $p, q \geq 1$, and let r be the odd one of p and q. Then*

$$N(a^p b^q) = \frac{n-1}{2} + \frac{r-1}{2}.$$

In particular $N(a^p b^q) \geq (n-1)/2$, with equality exactly for $a^{n-1}b$ and ab^{n-1}.

Proof. By complementation we may assume that p is even and q is odd. Fix m with $2m \leq n-1$ and consider the factor $W(i, m)$, of length $2m < n$. The circular word $a^p b^q$ has two run boundaries. If $W(i, m)$ contains no boundary it is a^{2m} (possible when $2m \leq p$) or b^{2m} (when $2m \leq q$), of types (m, m) and $(m, 0)$. If it contains exactly one boundary it is $a^s b^{2m-s}$ or $b^s a^{2m-s}$ with $1 \leq s \leq 2m-1$; its halves contain $\min(s, m)$ and $\max(s-m, 0)$ letters of the first kind, which are never equal, so it is not a square. If it contains both boundaries, it contains one of the two runs entirely, and since $2m < n$ not both. Suppose it contains the run b^q: $W(i, m) = a^s b^q a^{s'}$ with $s, s' \geq 1$ and $s + s' = 2m - q$. If $s \geq m$ the halves contain m and $s + s' - m = m - q < m$ letters a. If $s < m$ and $m - s > q$ the halves are $a^s b^q a^{m-s-q}$ and a^m, with $m - q < m$ letters a in the first. If $s < m$ and $m - s \leq q$ the halves are $a^s b^{m-s}$ and $b^{q-m+s} a^{s'}$, a square exactly when $s = s'$, that is $2s = 2m - q$, impossible since q is odd. Suppose it contains the run a^p: $W(i, m) = b^s a^p b^{s'}$ with $s, s' \geq 1$, $s + s' = 2m - p$. As before the halves have the same number of letters a only when $s < m$, $m - s \leq p$ and $m - s = p - (m - s)$, that is $s = s' = m - p/2$; this needs $m \geq p/2 + 1$, and then $s = s' \leq p$ because $m \leq (n-1)/2 = (p + q - 1)/2$ gives $m - p/2 \leq (q-1)/2 \leq q$ and also $s \leq q$; so for every m with $p/2 + 1 \leq m \leq (n-1)/2$ there is exactly one such type, $(m, p/2)$. Altogether the types of half-length m are: (m, m) when $m \leq p/2$; $(m, p/2)$ when $m \geq p/2 + 1$; and $(m, 0)$ when $m \leq (q-1)/2$. The first two lists provide one type for each $m \in [1, (n-1)/2]$, and the third provides $(q-1)/2$ further types, distinct from the others because $p \geq 2$. This gives the formula with $r = q$. Equality $N = (n-1)/2$ means $q = 1$. □

The alternating word $(ab)^{(n-1)/2}a$ also has $N = (n-1)/2$ (a computation, confirmed for $n \leq 25$), and for $n \leq 13$ every word with $N = (n-1)/2$ other than a^n and its complement has one letter isolated (there are 4, 7, 6, 7 such words for $n = 7, 9, 11, 13$ up to conjugation and complement). So, unlike the distinct count of Theorem 1.1, the minimum of N in odd length is attained by a large family of words. We expect it to be the same number.

Conjecture 5.5. Let n be odd. Every binary circular word w of length n satisfies $N(w) \geq (n-1)/2$; more precisely

$$z_1(w) + z_2(w) + \cdots + z_k(w) \geq k \qquad \text{for } 1 \leq k \leq \frac{n-1}{2}.$$

Both statements have been verified for every odd $n \leq 27$; Proposition 5.3 is the case $k \leq 2$, Proposition 5.4 the case of two runs. Together with Theorem 1.1 the conjecture says that for odd n the least number of non-equivalent

abelian squares in a circular word equals the least number of distinct ones, $(n-1)/2$, whereas [2] gives $\lfloor (n+1)/4 \rfloor$. The conjecture cannot be read as a statement about each half-length separately: *aababbbbb* has no square of half-length 3, and $a^{k+1}ba^kb^{n-2k-2}$ none of half-length m for $k+2 \le m \le (n-3)/2$ (a computation for $k \le 6$, $n \le 29$), the deficit being made up by the types $(m,0)$ and (m,m) of the long runs at small half-lengths. Nor does the inequality survive for even n: $z_1 = 0$ for $(ab)^{n/2}$, and $z_3 = z_4 = 0$ for $a^3b^3a^3b^5$, which has only three types of half-length at most four.

6. The other problems of the survey

Simpson's survey [1] lists six further problems on abelian squares. We record their status after a search of the literature up to October 2026, and we prove what we can about the last one.

6.1. **The linear minima.** Fici and Saarela conjectured that a binary word of length n contains at least $\lfloor n/4 \rfloor$ distinct abelian squares, with equality for $n = 4k+3$ only at $a^{2k+1}ba^{2k+1}$ and its complement, and Fraenkel, Simpson and Paterson conjectured the same bound for non-equivalent abelian squares, with the further extremal words $(ab)^{2k+1}a$ (Conjectures 0.4 and 0.5 of [1]). Fazekas, Mammoliti, Mercaş and Simpson [3] prove the bound for the words in which some letter occurs at most three times and extend it to Conjecture 4 of [3]: a word with exactly $\lfloor n/4 \rfloor$ distinct abelian squares contains only the squares a^{2j} and b^{2j}. The program `linear_minima.c` shows the following. For every $n \le 24$ the least number of distinct, and the least number of non-equivalent, abelian squares in a binary word of length n is $\lfloor n/4 \rfloor$. The equality clause of Conjecture 4 of [3] fails for $n = 4$, where *abab* has exactly $\lfloor 4/4 \rfloor = 1$ abelian square, which is not of the form a^{2j} or b^{2j} (the word appears in Table 1 of [3]); for $5 \le n \le 24$ the clause holds. We have no proof of the linear conjectures, and the circular results do not imply them: a linear word has at most the abelian squares of the circular word obtained by closing it, and closing $a^{2k+1}ba^{2k+1}$ gives $a^{4k+2}b$, with $D = 2k+1$ by Theorem 1.1, while the linear word has only k.

6.2. **Conjecture 0.1 (Fici and Mignosi).** The conjecture that the number of distinct abelian squares in a word of length n over any alphabet is at most the maximum over binary words is claimed in the preprint [9]; the argument given there is a sketch whose steps we could not complete, and we regard the conjecture as open.

6.3. **Conjecture 0.2 (Kucherov, Ochem and Rao).** The conjecture that a word of length n has $O(n^{3/2})$ non-equivalent abelian squares is open. Kociumaka, Radoszewski, Rytter and Waleń [10] prove that the maximum is $\Omega(n^{3/2})$ and $O(n^{11/6})$, and that the maximum number of distinct abelian squares is $\Theta(n^2)$. We have nothing to add.

6.4. **Conjecture 0.3 (Mäkelä).** The conjecture that some infinite ternary word has no abelian square other than *aa*, *bb*, *cc* is open. Rao and Rosenfeld [11] show that long abelian squares can be avoided over three letters and,

on the binary side, that abelian cubes of abelian period at least 2 cannot be avoided; see [6] for the state of the question. We have nothing to add.

6.5. **Question 0.7 (occurrences of abelian squares).** Let $T(n)$ be the least number of occurrences of abelian squares in a binary word of length n, where an occurrence is a pair (i, m) such that $x_i \cdots x_{i+2m-1}$ is an abelian square. Simpson asked for $T(n)$; his table, continued as sequence A268084 of [12], gives $T(n)$ for $n \le 60$, with $T(20) = 20$, $T(40) = 52$ and $T(60) = 96$. The program `total_occurrences.c` recomputes $T(n)$ for $n \le 28$ and agrees with the sequence.

Theorem 6.1. (i) *$T(n+n') \ge T(n) + T(n')$ for all $n, n' \ge 1$. Consequently $T(n)/n$ converges to $\tau = \sup_{n\ge1} T(n)/n \in [1.6, \infty]$.*

(ii) *$T(n) \le \sum_{m=1}^{\lfloor n/2 \rfloor} (n - 2m + 1)\binom{2m}{m} 4^{-m} < n^{3/2}$ for all $n \ge 1$.*

Proof. (i) Let w be a word of length $n + n'$ with $T(n + n')$ occurrences. Its prefix of length n and its suffix of length n' have at least $T(n)$ and $T(n')$ occurrences, and an occurrence inside the prefix is not an occurrence inside the suffix. By Fekete's lemma for superadditive sequences, $T(n)/n$ converges to $\sup_n T(n)/n$, which is at least $T(60)/60 = 1.6$.

(ii) Let w be a uniformly random binary word of length n. For $1 \le m \le n/2$ and $0 \le i \le n - 2m$ the numbers of letters a in $x_i \cdots x_{i+m-1}$ and in $x_{i+m} \cdots x_{i+2m-1}$ are independent binomial variables with parameters m and $1/2$, so the probability that (i, m) is an occurrence is $\sum_j \binom{m}{j}^2 4^{-m} = \binom{2m}{m} 4^{-m}$. The expected number of occurrences is the sum in (ii), and some word has at most the expected number. Finally $\binom{2m}{m} 4^{-m} \le (\pi m)^{-1/2}$ and $\sum_{m \le n/2} m^{-1/2} < 2\sqrt{n/2}$, so the sum is less than $n\sqrt{2n/\pi} < n^{3/2}$. □

The values of [12] suggest that $T(n)/n$ keeps growing; whether τ is finite, that is, whether some infinite binary word has only linearly many occurrences of abelian squares in its prefixes, is open. A positive answer would need words that are far from random, since a random word has about $0.53\, n^{3/2}$ occurrences.

7. Formal verification

Theorem 1.1 and Corollary 1.2 are formalized in Lean 4 [7] with Mathlib [8], version 4.34.1, and the formal proofs are checked by the Lean kernel. Words are lists over a two-letter alphabet, and the definitions follow Section 1 literally: an abelian square is a word uv with u nonempty and the same number of each letter in u and v; the factors of a circular word w of length n are the prefixes of length L, $1 \le L \le n$, of its n rotations; and $D(w)$ is the number of distinct factors that are abelian squares. The formal statements are the following. For every $n \ge 1$ the least value of D on the words of length n is $\lfloor (n-1)/2 \rfloor$. A word of odd length, and a word of even length, attains this value exactly when it is a conjugate of one of the words in (a), respectively (b), of Theorem 1.1. Part (a) of Conjecture 0.6 holds, and part (b), stated literally, is false. Finally $D(ababab) = 2$, and $ababab$ is not a conjugate of $a^k b^{6-k}$ or of its complement for any k. The repository verification of the pinned source checks

these statements; the verification record accompanying this edition identifies the commit and successful run. The proofs depend only on the axioms of Lean's logic, namely propositional extensionality, choice and the soundness of quotients; they contain no unproved step and add no axiom. The results of Sections 5 and 6 are not formalized.

Independent computations confirm Theorem 1.1 on every binary circular word of length at most 22 and on one word of each necklace class up to length 28, and they confirm each lemma of Sections 2 and 3 on every word of length at most 16.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Claude (Anthropic) assisted with the original proofs, Lean formalization, computations and manuscript preparation. OpenAI Codex assisted with the source audit, small-case verification and conversion to the amsart format for this edition. The mathematical proof body is retained. The author is responsible for the content; no AI system is an author.

Data availability

Deep Bhattacharjee computed the Lean formalization and the Julia, C, C++, shell and Python computations with the help of AI assistance (Claude, Anthropic). The Lean formalization of the proofs in this paper and the programs are in release v2.3.0 of the repository `https://github.com/creelie/literate-octo-lamp`, `https://github.com/creelie/literate-octo-lamp/releases/tag/v2.3.0`; the programs of Sections 5 and 6 are the ancillary files `odd_circular_types.c`, `linear_minima.c` and `total_occurrences.c`; the release is archived on Zenodo as 10.5281/zenodo.23692500; the earlier preprint is Hexagon 2610.00473v1.

Competing interests

None.

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